National mortality rates from chronic liver disease and consumption of alcohol and pig meat.

A correlation between national pig-meat consumption and mortality rates from chronic liver disease (CLD) across developed countries was reported in 1985. One possible mechanism explaining this may be hepatitis E infection spread via pig meat. We aimed to re-examine the original association in more recent international data. Regression models were used to estimate associations between national pig-meat consumption and CLD mortality, adjusting for confounders. Data on CLD mortality, alcohol consumption, hepatitis B virus (HBV) and hepatitis C virus (HCV) seroprevalence for 18 developed countries (1990-2000) were obtained from WHO databases. Data on national pig-meat and beef consumption were obtained from the UN database. Univariate regression showed that alcohol and pig-meat consumption were associated with mortality from CLD, but beef consumption, HBV and HCV seroprevalence were not. A 1 litre per capita increase in alcohol consumption was associated with an increase in mortality from CLD in excess of 1.6 deaths/100,000 population. A 10 kg higher national annual average per capita consumption of pork meat was associated with an increase in mortality from CLD of between 4 and 5 deaths/100,000 population. Multivariate regression showed that alcohol, pig-meat consumption and HBV seroprevalence were independently associated with mortality from CLD, but HCV seroprevalence was not. Pig-meat consumption remained independently associated with mortality from CLD in developed countries in the 1990-2000 period. Further work is needed to establish the mechanism.